

Diogo de Bastos 🙌

Lisbon, Portugal | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

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Summary 📖

Lead AI/ML Engineer (5+ years) with a Ph.D. in Physics and a track record of deploying machine learning solutions that operate on large-scale, high-velocity data. I pair rigorous statistical modelling with production engineering, working closely with product, engineering, and risk stakeholders to deliver resilient analytics. Recent focus spans LLM-enabled automation, agentic systems, and platform-level AI engineering, alongside time-series forecasting and optimisation, with experience maintaining distributed compute environments within global collaborations.

Work experience 🏢

EY-Parthenon

Lisbon, 2026/05 - current | Lead AI/ML Engineer, AI Hub (Europe West) EY-Parthenon

- Hands-on builder within EY-Parthenon's Europe West AI Hub, shipping agentic systems, LLM integrations, and data pipelines at the platform level, infrastructure that dozens of teams and applications depend on across the region.
- Built and rolled out the firm's Model Context Protocol (MCP) server, taking it from prototype to a supported internal service and growing adoption from 0 to 100+ users across consulting teams.
- Designed and deployed production agents that automate research, document analysis, and client deliverables, including tool use, retrieval, evaluation harnesses, and guardrails for reliability and cost control.
- Applied causal inference (difference-in-differences, synthetic control, uplift modelling) to quantify treatment effects where experimentation is impossible, giving clients defensible estimates of what actually drove an outcome.
- Built Bayesian models for parameter estimation and forecasting under sparse data, propagating full posterior uncertainty into decisions so recommendations come with credible intervals rather than point estimates.
- Lead AI-first teams of 3-5 engineers, owning technical direction, hiring and onboarding, code and model review, and the delivery cadence from scoping to production handover.

EDP

Lisbon, 2022/09 - 2026/05 | Senior data scientist, AI/ML & quantitative analyst

- Reduced short-term price forecast error from 15% to 5% by combining LSTM, CNN, and Temporal Fusion Transformer models implemented in Python with PyTorch and SQL-backed data pipelines.
- Designed portfolio optimisation and algorithmic trading frameworks that improved gross margin by 30%, integrating ML outputs into decision workflows for trading and risk teams.
- Migrated analytics from MATLAB and R to a Python ecosystem, formalising code efficiency, scalability, and reproducibility for globally distributed stakeholders.
- Deployed predictive services and analytical applications on Databricks Services to meet availability, monitoring, and scalability requirements.
- Built LLM-powered assistants using Azure OpenAI, Llama2, Retrieval Augmented Generation, Weaviate, and LangChain to automate report generation and data interpretation. PoCs for report automation and research summarisation. Cut manual review time by >40%.

- Mentored colleagues on MLOps practices, supervised a master's thesis on deep learning for price forecasting, and acted as product lead for ML, optimisation, and trading initiatives.

Whitesmith

Coimbra, 2016/01 - 2017/06 | Business and hardware development

- Co-founded a cold-chain IoT startup, managing product strategy, client acquisition, and data-driven validation of market needs.
- Built sensor-to-cloud monitoring across 24 devices, integrating wireless communications (2.1 GHz/433 MHz), embedded prototyping (Arduino, Raspberry Pi), and analytics (SQLite, MATLAB).
- Designed low-power PCBs with Altium and coordinated with engineers and customers to deliver reliable hardware deployments.

jeKnowledge

Coimbra, 2014/09 - 2016/02 | CEO

- Expanded the non-profit tech organisation from 10 to 30 members by redesigning recruitment and leadership development.
- Directed rebranding, launched a summer academy focused on programming and hardware, and created networking events linking academia with startups.
- Oversaw hackathons and community programs, honing stakeholder communication and execution under tight timelines.

Research

Ph.D.

2018/01 - 2023/07 | High Energy Physics and Quantum Machine Learning

- Advanced supersymmetry searches within the CMS experiment by building C++ frameworks, boosted decision trees (ROOT TMVA), and TensorFlow neural networks to classify proton-proton collision data.
- Performed Bayesian statistical analyses to set stringent limits on top squark pair production cross sections at low masses, managing class imbalance and detector noise.
- Experimented with Variational Quantum Classifiers and Quantum GANs (Python, PennyLane, Qiskit) to accelerate Monte Carlo data augmentation.
- Tested and calibrated ASICs for the CMS Timing Detector upgrade and maintained computing operations for the CMS collaboration, ensuring data integrity across distributed resources.
- Mentored junior researchers and organised outreach events to promote STEM education.

Education

- Doctorate of Philosophy in Physics, Instituto Superior Técnico Universidade de Lisboa - Thesis: "Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at the Large Hadron Collider (Pass with Distinction)"
- Master's in Physics Engineering, University of Coimbra - Thesis: "Automated monitoring and diagnosis of cold chains" (18/20)

Publications

- 2023/06/12 - The CMS Collaboration, "Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at the Large Hadron Collider" [JHEP06\(2023\)060](#)

- Bastos, D. "Using Variational Quantum Algorithms and Quantum Generative Adversarial Networks for Supersymmetry in High-Energy Physics" (internal paper)
- The CMS Collaboration, "Experimental characterization of the BTL Front-end Board based on TOFHIR1" (internal note)
- Speaker: "Predicting Spain power price with Deep Learning," 11th Annual Electricity Price Forecasting and Modelling Forum, 2024.
- Speaker: "Searches for top squarks in compressed scenarios with the CMS experiment," SUSY Conference, 2022.

Skills

- Programming: Python, C++, R, MATLAB, Bash
- Data Analysis & Visualisation: Pandas, NumPy, Matplotlib, Seaborn
- Machine Learning: PyTorch, TensorFlow, scikit-learn, Keras, neural networks, LSTM, TFT, CNN, BDT, genetic programming, MLOps, data pipelines
- NLP & GenAI: Azure OpenAI, Llama2, LangChain, Retrieval Augmented Generation, agentic systems / AI agents, LLM integration, Claude Code, Cloudflare Workers AI, speech-to-text, translation, text-to-speech
- Databases & Platforms: SQL, MySQL, Oracle, Azure, Databricks
- Statistical & Optimisation Techniques: Quadratic programming, risk control, Bayesian analysis
- Leadership & Communication: Portuguese and English fluency, mentorship, outreach, conference speaking

Conferences

- 2024/09/13 - Speaker: "Predicting Spain power price with Deep Learning", 11th Annual Electricity Price Forecasting And Modelling Forum
- 2022/06/29 - Speaker on behalf of the CMS collaboration: "Searches for top squarks in compressed scenarios with the CMS experiment", The XXIX International Conference on Supersymmetry and Unification of Fundamental Interactions, Ioannina, Greece
- 2020/06/26 - Speaker: "High-performance timing detector for the HL-LHC Upgrade of the CMS experiment at CERN", 6th IDPASC /LIP PhD student Workshop, remote
- 2020/03/09 - Seminar: "Presentation to the CAT", Lisbon, Portugal
- 2020/02/15 - Speaker: "CMS: Searching for stop", Braga, Portugal
- 2019/09/25 - Invited speaker: "Distributed Computing at the CMS Experiment: From the point of view of a physicist", IBERGRID 2019, Santiago de Compostela, Spain
- 2019/07/01 - Speaker: "Search for the SuperSYmmetric partner of the top quark at the LHC with a multivariate approach", 5th IDPASC /LIP PhD student Workshop, Braga, Portugal
- 2019/04/06 - Invited mentor: "IST Masterclasses", Lisbon, Portugal
- 2019/02/11 - Invited speaker: "Particles - from the Universe to the lab", Lisbon, Portugal
- 2017/10/26 - Organizer of [LISBON.AI](#), a conference dedicated to Artificial Intelligence for 150 engineers
- 2015/07/23 - Organizer of [Summer JADE Meeting](#), a 4-day international congress for 300+ Entrepreneurs